Shahjalal University of Science and Technology

Department of Electrical and Electronic Engineering
3rd Year 2nd Semester Final Examination, 2023
Course Code: EEE 329, Session: 2020-2021
Course Title: Basic Communication Engineering

Total Marks: 100 **Credits: 3.0** **Total Time: 3 Hours**

Part-A

[Answer any two of the following questions]

1. (a) Find the entropy of single bit binary memoryless source. Also draw the curve of entropy Vs. probability for that source. 5

   (b) Briefly explain the three major bands of the electromagnetic (EM) spectrum-radio waves, microwaves and infra-red waves-in terms of their bandwidth, propagation mode and applications. 6

   (c) Write short note on-

   (i) Thermal Noise
   (ii) Transit Time Noise 4

   (d) A carrier wave of frequency 10 MHz and peak value of 10 V is amplitude modulated by a 5 kHz sine wave of amplitude 6 V. Determine the modulation index, side frequencies and amplitude of side frequencies. 4

   (e) An audio frequency signal $10\sin(2\pi 500t)$ is used to amplitude modulated a carrier of $50\sin(2\pi 105t)$ [*sic*: should be $2\pi 10^5 t$]. Calculate:

   (i) Modulation index
   (ii) Side-band frequencies
   (iii) Total power delivered to the load of 600 Ω 6

2. (a) Differentiate between multiplexing and multiple access technologies. 5

   (b) Five channels, each with 100 kHz bandwidth, are to be frequency division multiplexing together. What is the minimum bandwidth of the link if there is a need for a guard band of 10 kHz between the channel to prevent interference? 5

   (c) With proper block diagram explain the relation between frequency modulation and phase modulation. 5

   (d) In an FM system, when the audio frequency is 500 Hz and modulating voltage 2.5 V, the deviation produced is 5 kHz. If the modulating voltage is now increased to 7.5 V, calculate the new value of frequency deviation produced. If the AF voltage is raised to 10 V while the modulating frequency dropped to 250 Hz, what is the frequency deviation? Calculate the modulation index in each case. 10

3. (a) Define slope overload distortion and granular noise in delta modulation and how it is removed in ADM? 5

   (b) Draw the transmitter and receiver circuits block diagram of delta modulation. 5

   (c) Explain the basic block diagram of PCM system. 5

   (d) An analog signal having 4 kHz bandwidth is sampled at 1.25 times the Nyquist rate, and each sample is quantized into one of 256 equally likely level. Assuming the samples to be statistically independent:

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(i) What is the information rate of the source?
(ii) Can the output of this source be transmitted without error over the AWGN channel with a bandwidth of 10 kHz and S/N ratio of 20 dB?
(iii) Find S/N ratio required for error-free transmission of part ii.
(iv) Find the bandwidth required for an AWGN channel for an error free transmission of output of this source if S/N ratio is 20 dB 10

Part-B

[Answer any two of the following questions]

4. (a) With appropriate block diagram, describe the operation of superheterodyne receiver. 7
(b) Explain how the message signal is reconstructed from VSB signal. 7
(c) State and prove Nyquist sampling theorem. Also find the interpolation formula for the reconstruction of analog signals from its uniform samples. 7
(d) State Carson's rule and its limitations. 4

5. (a) Write down the comparison between TDMA and CDMA. 5
(b) A voice grade telephone channel has a bandwidth of 3400 Hz. If the signal to noise ratio (SNR) on the channel is 30 dB, determine the capacity of the channel. If the above channel is to be used to transmit 4.8 kbps of data, determine the minimum SNR required on the channel. 6
(c) It is desired to set up a central station for simultaneous monitoring of the electrocardiograms (ECGs) of 10 hospital patients. The data from the 10 patents [*sic*: should be "patients"] are brought to a processing center over wires and are sampled, quantized, binary-coded, and time-division-multiplexed. The multiplexed data are now transmitted to the monitoring station. The ECG signal bandwidth is 100 Hz. The maximum acceptable error in sample amplitude is 0.25% of the peak signal amplitude. The sampling rate must be at least twice the Nyquist rate. Determine the minimum cable bandwidth needed to transmit these multiplexed data. 7
(d) Consider a message signal m(t) with the spectrum shown in Fig. 1. The message bandwidth W = 1 kHz. This signal is applied to a product modulator, together with a carrier wave $A_c \cos(2\pi f_c t)$, producing the DSBSC modulated wave s(t). This modulated wave is next applied to a coherent detector. Assuming perfect synchronism between the carrier waves in the modulator and detector, determine the spectrum of the detector output when:

(i) the carrier frequency $f_c = 1.25$ kHz and
(ii) the carrier frequency $f_c = 0.75$ kHz.

What is the lowest carrier frequency for which each component of the modulated wave s(t) is uniquely determined by m(t)? 7

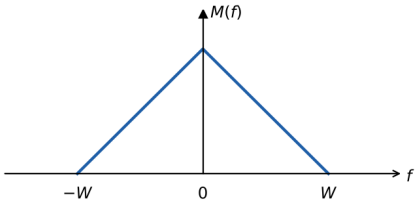


Fig. 1 of the paper (redrawn): the message spectrum M(f)

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6. (a) Briefly describe FSK generation with the necessary block diagram. Include its spectrum and discuss the bandwidth requirements, along with the advantages and disadvantages of FSK. 8

   (b) With proper block diagram explain the principle of coherent detection of QAM. Mention its advantages and drawbacks. 7

   (c) Explain the principle of frequency hopping spread spectrum technique. 6

   (d) Tabulate the values of signal at each stage of DPSK encoder and receiver for message signal '1010011100' 4

Term Test 1 (its questions 1 to 4 are 7(a) to 7(d))

[EEE 329 (2020-21), Total Marks: 4 × 5 = 20, Total time: 40 minutes]

7. (a) Find the entropy of single bit binary memoryless source. Also draw the curve of entropy vs probability for that source. 5

   (b) Explain distortion as a transmission impairment in communication systems. What are the conditions required for distortionless transmission? 5

   (c) We have a communication channel with a bandwidth of 5 MHz. A signal with an amplitude of 10V is transmitted, and the receiver detects a signal with an amplitude of 2V and noise with an amplitude of 5 mV.

   (1) Calculate the attenuation.
   (2) Determine the maximum channel capacity.
   (3) To achieve a data rate of 100 Mbps, find the required signal level. 5

   (d) Find the equivalent noise resistance for several amplifiers in cascade. 5

Term Test 2 (its questions 1 to 5 are 8(a) to 8(e))

[EEE 329 (2020-21), Total Marks: 4 × 5 = 20 (Answer any 4), Total time: 40 minutes]

8. (a) Amplitude spectrum of a message signal shown in figure-1. Draw the spectrum of its DSB-WC and SSB-SC modulated signal. 5

   (b) Explain how phase and frequency error affect the coherent detection of AM. 5

   (c) The antenna current of an AM transmitter is 8 Amps, when only the carrier is sent, but it increases to 8.93A, when the carrier is modulated by a single sine wave. Find percentage modulation. Determine the antenna current when the percentage modulation changes to 0.8 5

   (d) Explain principle of preemphasis and deemphasis 5

   (e) Consider an FM signal obtained from a modulating signal frequency of 2000 Hz and maximum Amplitude of 5 volts. The frequency sensitivity of modulator is 2 kHz/V. Find the bandwidth of the FM using Carson's rule. 5

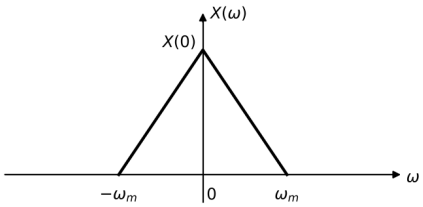


Figure-1 of the paper (redrawn): the amplitude spectrum X(ω) of the message

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